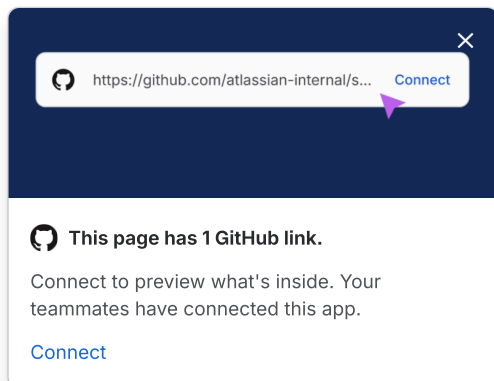


High level design- Compliance bot

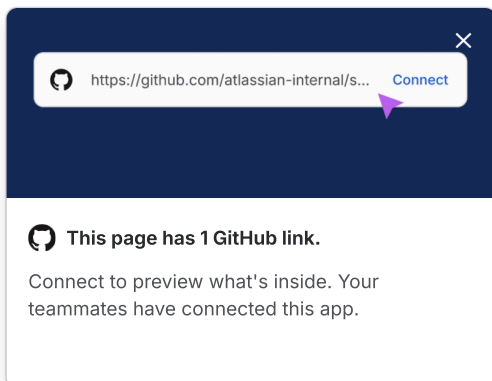
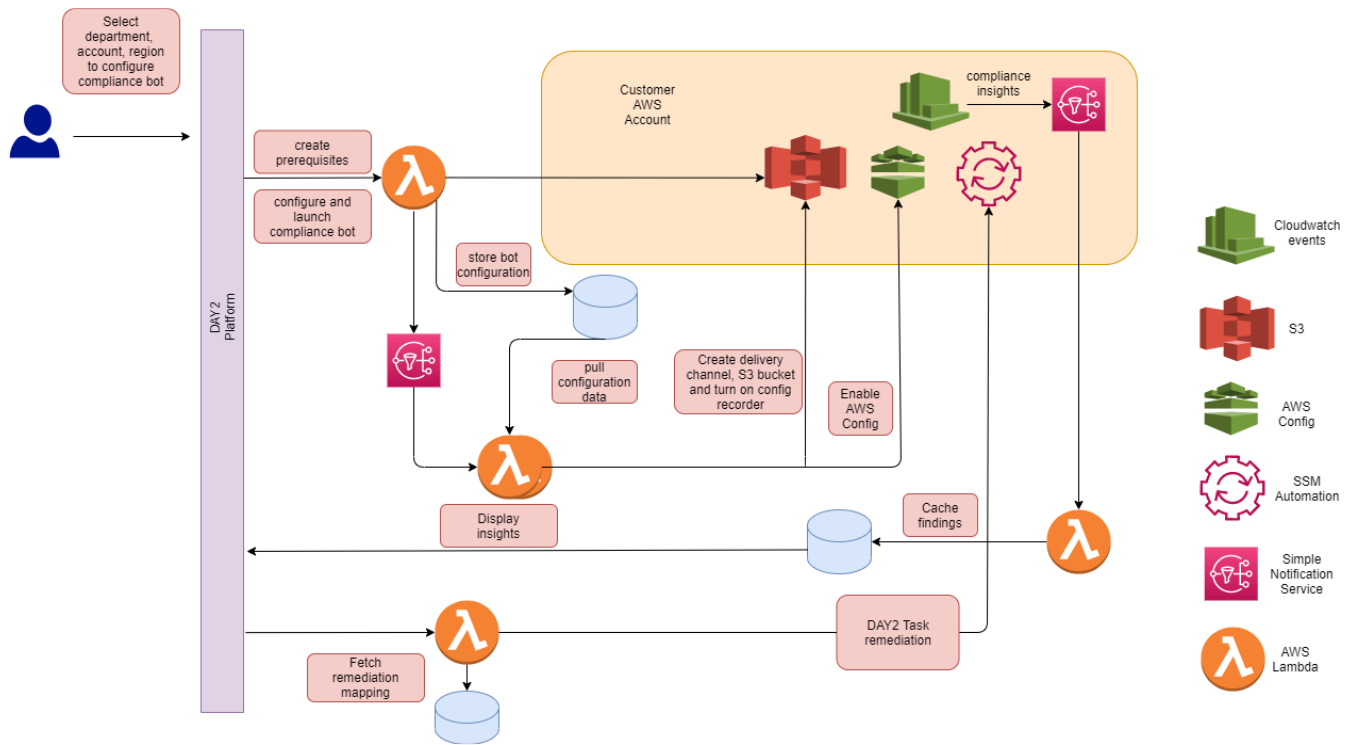
- [Overview:](#)
- [Architecture Diagram:](#)
- [Bot flow overview:](#)
- [Use cases:](#)
- [Database:](#)
 - [ComplianceInsights](#)
 - [Bots](#)
 - [Bot configuration](#)
 - [Bot instances](#)
 - [Policy Detail](#)
- [API:](#)
- [Events:](#)
- [Open questions:](#)

Overview:

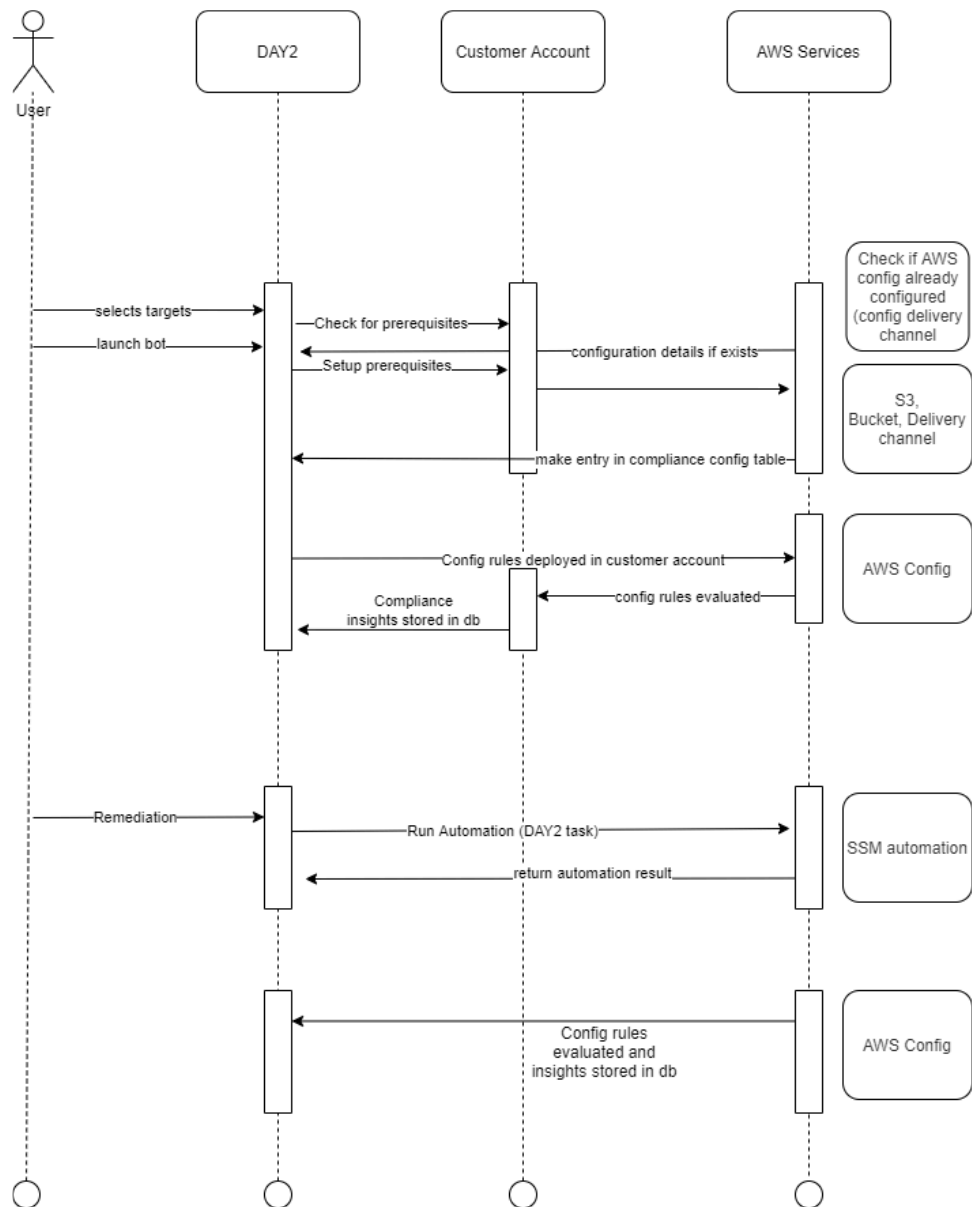
- Describes the high-level design of compliance bot.
- Goal: As a customer, I would like to have automated detection and response for **Compliance** Industry Standards and AWS best practices so I could achieve **Compliance** for my cloud footprint via DAY2 Platform.
- To achieve this behavior we would be using BOT Framework which will be deployed on the customer's account and will provide us Compliance status of different resources based on some pre-defined rules.
- AWS Config service will be used to get the compliance status.
- BOT will behave as a service which will be scheduled to perform its task and go back to sleep state.
- BOT can be on-boarded, updated and off-boarded.



Architecture Diagram:

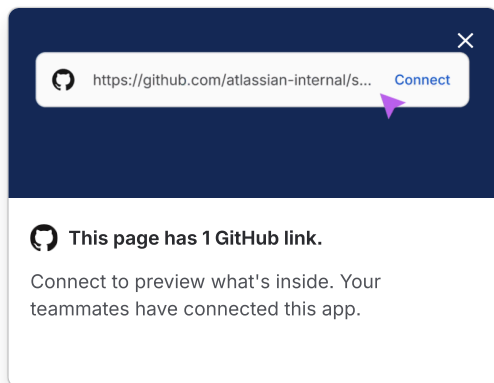


Bot flow overview:



Use cases:

1. Register bot to Day2



olution with automated detection and response for standards and AWS best practices.

eter definition

1. Department

2. Account

3. Region

3. **Get bot parameters:** Get bot parameters

4. **Bot policy rule remediation mapping registry**

- a. Policy - S3 best practices
- b. Rule - AWS Config Rule (S3BucketPublicWriteProhibited)
- c. Remediation - AWS-DisableS3BucketPublicReadWrite (Automation document provided by AWS)
- d. Remediation Type - Day2 Task

5. **Configure bot**

- a. Configuration Bot will include storing all the configuration details required to launch the ComplianceBot at one or more targets.
 - i. Account and regions list on which ComplianceBot needs to be enabled.
 - ii. AWS Config rules that have to be deployed as selected by the user.

6. **Launch bot in a target**

7. **Validate prerequisites in a target**

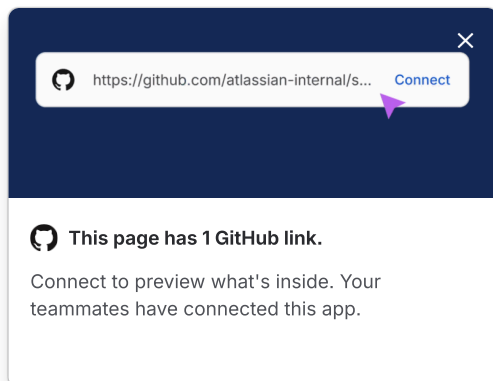
- a. In order to validate if AWS Config is already turned on, the steps mentioned in the following link will be used ([AWS Document to verify prerequisites.](#))
 - i. Verify s3 bucket is created
 - ii. Verify IAM role is created
 - iii. Verify SNS topic is created
 - iv. Verify that the Delivery Channel Is Created
 - v. Verify that the Configuration Recorder Is Created
 - vi. Verify that AWS Config has started recording
- b. If a region already has AWS config turned on we will not change the config settings, Only the config rules will be deployed in this region.

NOTE: In this case we would need to store this info that pre-requisites configs are owned and existing configurations of user will not get deleted.

If AWS Config in a region, the following things are required:

1. IAM role (IAM role that grants AWS Config permission to record and send this information to Amazon S3 and Amazon SNS)

2. S3 bucket (one will be created per account config history of all regions)



- Creating a Cloudwatch event rule (MontyCloud-TaskNotificationTopic will be the rule target)
- Use API's (**put_delivery_channel** and **put_configuration_recorder**) to create a delivery channel and turn on config recorder in the specific region where the Compliance bot needs to be configured.
- Turn on AWS config: create recorder, create delivery channel
(NOTE: You can have one configuration recorder and one delivery channel per region in your account.)

9. Receive insights from bot instance

a. AWS Config sends notifications for the following events:

Among following list of notifications we would be storing ComplianceChangeNotification as insights.

- Configuration item change for a resource.
- Configuration history for a resource was delivered for your account.
- Configuration snapshot for recorded resources was started and delivered for your account.
- Compliance state of your resources and whether they are compliant with your rules.
- Evaluation started for a rule against your resources.
- AWS Config failed to deliver the notification to your account.

b. A sample of the compliance message received from the SNS topic

```

1 {
2   "awsAccountId": "717930997628",
3   "configRuleName": "S3BucketPublicReadProhibited-conformance-pack-gsfzeeqwo",
4   "configRuleARN": "arn:aws:config:us-east-1:717930997628:config-rule/aws-service-
   rule/config-conforms.amazonaws.com/config-rule-g3dyxq",
5   "resourceType": "AWS::S3::Bucket",
6   "resourceId": "cf-templates-1coep36ivvzq3-us-east-1",
7   "awsRegion": "us-east-1",
8   "newEvaluationResult": {
9     "evaluationResultIdentifier": {
10      "evaluationResultIdentifier": {
11        "S3BucketPublicReadProhibited-conformance-pack-gsfzeeqwo",
12        "AWS::S3::Bucket",
13        "cf-templates-1coep36ivvzq3-us-east-1"
14      },
15      "timestamp": "2020-10-19T14:49:29.904Z"
16    },
17    "compliance": "COMPLIANT",
18    "timestamp": "2020-10-19T14:49:42.156Z",
19    "start": "2020-10-19T14:49:41.809Z",
20    "end": "2020-10-19T14:49:42.156Z"
21  },
22   "oldEvaluationResult": null,
23 }
```


<https://github.com/atlassian-internal/s...>
Connect



This page has 1 GitHub link.

Connect to preview what's inside. Your teammates have connected this app.

```

24 "notificationCreationTime": "2020-10-19T14:49:42.324Z",
25 "messageType": "ComplianceChangeNotification",
26 "recordVersion": "1.0"
27 }

```

NOTE: For MVP we will be caching only “ComplianceChangeNotification” insights on the platform side.

1. There are 2 things based on which we can check compliance by using AWS Config:

- **Config Rules:**

- These are of two types AWS managed rules and Custom rules.
- AWS-managed rules are provided by AWS most common scenarios are available [List of AWS config managed rules](#) and they can also be customized.
- Eg: ec2-managedinstance-applications-required : Checks whether all of the specified applications are installed on the instance. Optionally, specify the minimum acceptable version. Optionally, specify the platform to apply the rule only to instances running that platform.
- AWS Config evaluates resources based on triggers these triggers can be 1) Change in configuration of the resource 2) Runs on the frequency that you choose.

- **Conformance packs:**

- A conformance pack is a collection of AWS Config rules and remediation actions that can be easily deployed as a single entity in an account and a Region.
- As of now, we are going ahead with the packs provided by AWS.
- Execution Interval of these packs cannot be changed and will be based on the frequency defined in each rule within the pack.

2. **Deploying Config rules.**

- **put_config_rule** API will be used to deploy the config rules to the account.

NOTE: For MVP we will be supporting only AWS-Managed Config rules.

Eg: “[ec2-managedinstance-applications-required](#)” managed config rule for server compliance which checks whether the specified applications are installed on the instance .

aws-config-rules

Conformance packs has the parameters with default values, for MVP deployment with default values and not user input.

aws-config-rules/blob/master/aws-config-conformanc

practices-for-ACSC-ISM.yaml

Connect your Github account

This page has 1 GitHub link.

Connect to preview what's inside. Your teammates have connected this app.

- If customer account already has config rules deployed, newly deployed config rules may raise limit exceed per account.

There is no API to get count of deployed config rules but we can calculate number of config rules deployed with `describe_config_rules` API

- config rules are region specific

e.g [Operational Best Practices for ABS CCIG 2.0 Material Workloads](#)

AWS Region: All supported AWS Regions except Asia Pacific (Hong Kong), Europe (Stockholm), and Middle East (Bahrain)

3. Scheduling:

- Evaluation frequency of each config rule is different and for most of the rules default is 24 hours
 - We can also allow the user to customize it while deploying the rules : `One_Hour` | `Three_Hours` | `Six_Hours` | `Twelve_Hours` | `TwentyFour_Hours` .
- NOTE:** For MVP we will not be allowing the customers to select evaluation frequency.
- Evaluation frequency will be based on the default values specified by AWS for each rule.

4. Getting results once the config rules are evaluated:

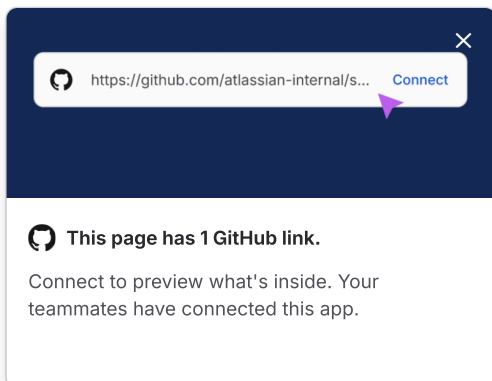
- when resources are evaluated against config rules and if there is `ComplianceChangeNotification` ([or Notifications listed](#)) we get a notification from the SNS topic with all the required information.

5. Store bot insights

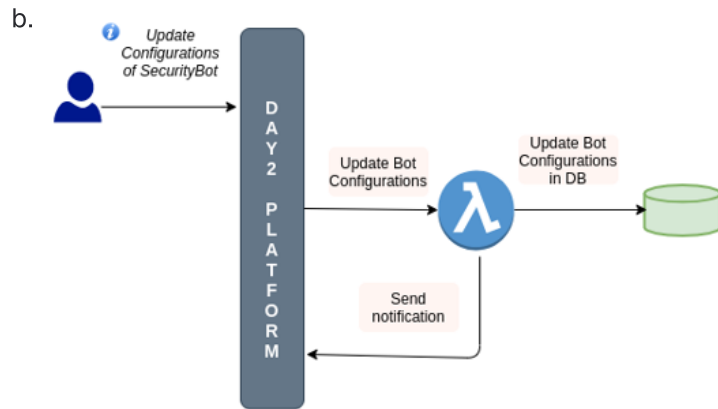
- The compliance change notifications that we receive through the SNS topic are then stored in the `ComplianceInsights` table.

6. Get bot data

- Get compliance insights API will be used to get the Insights data from



led, the user can go back and deploy more rules to a
config ruled from the target



8. Remediation

- a. AWS Config remediation for config rules is supported by using Systems Manager Automation documents.
- b. Based on the compliance insights data that we received and have stored in db we can give the user an option to remediate it.
- c. Approach:
 - i. Remediation will be triggered through DAY2 platform once the config rules are evaluated.

NOTE: For MVP we will be giving only the manual remediation option.

 - ii. We will provide the user with an option to remediate by manually selecting resources.
 - iii. By handling remediation from the platform side we have more control over when the remediation has been triggered and the status of the remediation action, whether it was a failure or success.

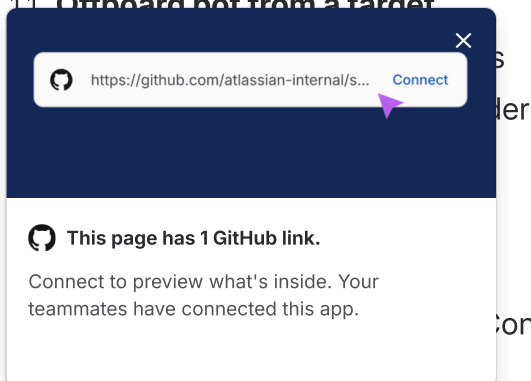
9. Trigger approval workflow

- a. Approval
 - i. Approval in the Day2 (Day2 users)
 - ii. Approval using SSM Automation document (IAM users)

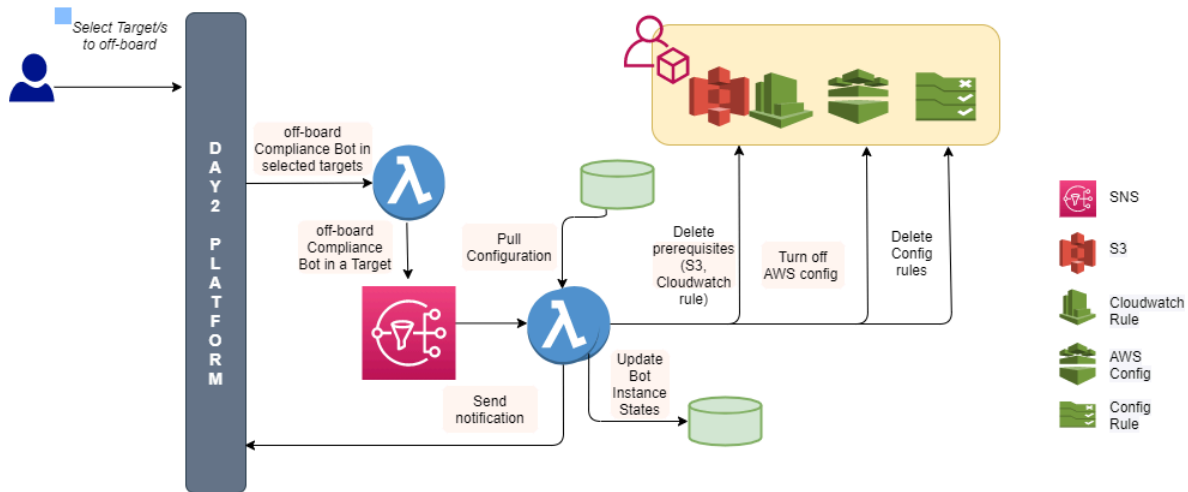
10. Capture health of a bot instances

- a. What should be the strategy for capturing the bot's health

11. Offboard bot from a target



config rule and all of its evaluation results)



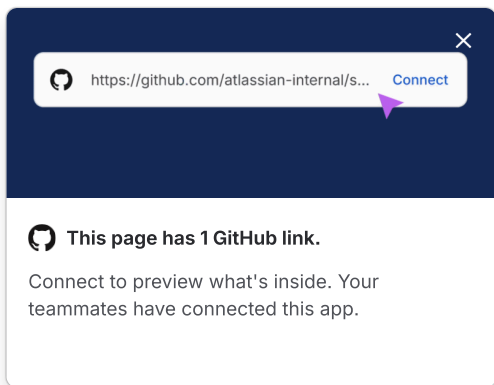
12. Bot Instance state

- a. Setup updates (Configuration: success or failed)

13. Bot executions update

14. Getting the latest versions of applications

- a. We will be checking for only 2 applications AmazonCloudwatchAgent and SSM Agent.
- b. We have to keep track of when the latest versions of these applications are available and keep them updated in our DB.
- c. Two different config rules will be created based on the OS type (one for windows and one for Linux)
- d. SSM Agent version updation:
 - i. Amazon Simple Notification Service (Amazon SNS) can notify you when new versions of SSM Agent are released. The steps to be followed are:
 1. Open the Amazon SNS console.
 2. From the Region selector in the navigation bar, choose **US East (N. Virginia)**, if it is not selected already. You must select this Region because the SNS notifications for SSM Agent that you are subscribing to are generated from this Region only.
 3. In the navigation pane, choose **Subscriptions**.
 4. Choose **Create subscription**.



on, do the following:

the following Amazon Resource Name (ARN):

1:720620558202:SSM-Agent-Update

AWS Lambda .

SNS task notification topic.

created and confirmed a lambda will get triggered by this

a new version is released and then we can update our db

with the latest version.

e. CloudWatchAgent version updation:

- i. There is no event or API which can give us the latest version of CW agent we have to visit the github link and then check the version and update it in our db.

Database:

ComplianceInsights

This table stores the evaluation results of config rules. (events that we receive through SNS)

ComplianceChangeNotification : When AWS Config evaluates your resources against a custom or managed rule, AWS Config sends a notification that shows whether the resources are compliant against the rule.

▼ Example Compliance Change Notification

```
1  {
2    'awsAccountId': '257105873514',
3    'configRuleName': 'cloud-trail-cloud-watch-logs-enabled',
4    'configRuleARN': 'arn:aws:config:us-east-1:257105873514:config-rule/config-rule-
5    p1j8wf',
6    'resourceType': 'AWS::CloudTrail::Trail',
7    'resourceId': 'MC-CloudTrail-Log',
8    'awsRegion': 'us-east-1',
9    'newEvaluationResult': {
10     'evaluationResultIdentifier': {
11       'evaluationResultQualifier': {
12         'configRuleName': 'cloud-trail-cloud-watch-logs-enabled',
13         'resourceType': 'AWS::CloudTrail::Trail',
14         'resourceId': 'MC-CloudTrail-Log'
15       },
16       'orderingTimestamp': '2020-10-30T14:00:41.776Z'
17     },
18     'complianceType': 'NOT_APPLICABLE',
19     'resultRecordedTime': '2020-10-30T14:01:01.046Z',
20     'configRuleInvokedTime': '2020-10-30T14:01:00.535Z',
21     'annotation': None,
22     'resultToken': None
23   },
24   'oldEvaluationResult': {
25     'evaluationResultIdentifier': {
26       'evaluationResultQualifier': {
27         'configRuleName': 'cloud-trail-cloud-watch-logs-enabled',
28         'resourceType': 'AWS::CloudTrail::Trail',
29         'resourceId': 'MC-CloudTrail-Log'
30       },
31       'orderingTimestamp': '2020-10-29T14:00:36.901Z'
32     },
33     'complianceType': 'NON_COMPLIANT',
34     'resultRecordedTime': '2020-10-29T14:02:58.709Z',
35     'configRuleInvokedTime': '2020-10-29T14:02:58.385Z',
36     'annotation': 'CloudTrail trail is not associated with any CloudWatch Logs',
37     'resultToken': None
38   }
39 }
```

 <https://github.com/atlassian-internal/s...> [Connect](#)

 This page has 1 GitHub link.

Connect to preview what's inside. Your teammates have connected this app.

```

37 },
38 'notificationCreationTime': '2020-10-30T14:01:01.903Z',
39 'messageType': 'ComplianceChangeNotification',
40 'recordVersion': '1.0'
41 }

```

Column	Type	Description
CustomerId	UUID	
AccountNumber	String(45)	
Region	String	Region of the config rule
ConfigRuleName	String(255)	Name of aws config rule
ComplianceStatus	String(45)	Compliance status for the rule COMPLIANT or NON-COMPLIANT
ResourceType	String(45)	Type of the resource
ResourceId	String(255)	Id of the resource
ResultExecutionTime	datetime	
lastEvaluationResult	String(255)	COMPLIANT or NON-COMPLIANT
CreatedAt	datetime	

Bots

Id	Name	Description	Manifest	Create At	UpdatedAt
uuid	botna	bot desc	<pre> 1 { 2 "BotId": "BotId", 3 "Name": "<BotName>", 4 "Description": "<Description>", 5 "Metadata": { 6 "Name": "<BotName>", 7 "Description": "<Description>", 8 "Avatar": "<Avatar>" 9 }, 10 "BotParameters": { 11 "InputParameters": [12 { 13 "Name": "Account" 14 } </pre>		

 <https://github.com/atlassian-internal/s...> [Connect](#)

 This page has 1 GitHub link.

Connect to preview what's inside. Your teammates have connected this app.

```
15      {
16        "Name": "Region"
17      }
18    ],
19  },
20  "BotPolicies": [
21    {
22      "PolicyGroupName": "
23      <PolicyGroupName>",
24      "PolicyGroupDescription": "
25      <PolicyGroupDescription>",
26      "PolicyGroupId": "
27      <uuid(PolicyGroupName)>",
28      "Policies": [
29        {
30          "PolicyName": "
31          <PolicyName>",
32          "PolicyDescription": "
33          <PolicyDescription>",
34          "PolicyId": "
35          <uuid(PolicyName)>",
36          "InputType": "<Text|
37          TextArea| List| CheckBox>"
38        }
39      ]
40    }
41  ],
42  "BotPermissions": {
43    "ResourceName": "
44    <ResourceName>",
45    "ResourceType": "
46    <ResourceType>"
47  },
48  "BotNotifications": [
49    {
50      "NotificationType": "
51      <Slack>",
52      "ChannelName": "
53      <ChannelName>",
54      "Type": "<COMPLIANT| NON-
55      COMPLIANT| ALL>"
56    }
57  ]
58 }
```

Bot configuration

This table will store the default configuration of the bot (common table for all the bots)

 <https://github.com/atlassian-internal/s...> [Connect](#)

 This page has 1 GitHub link.
Connect to preview what's inside. Your teammates have connected this app.

changes the global configuration that will be set as the default(launched after changing the configuration. Previously order config version (V1).

Configuration	UpdatedAt	UpdatedBy	CustomerId	Status
				aggregated status from

					insights
--	--	--	--	--	----------

Bot instances

This table is used to store bot instance configurations and their targets.

Bot ID	Bot instance ID	Bot target type	Bot target	Bot Configuration	DeploymentStatus	Custom erId	UpdatedAt	Instance Metadata
ComplianceBotId	ComplianceBotInstance1	AccountRegion	A1 R1	Configuration V1 { This will include all the selections done by user (also config rules selected)	[{ }] These will be updated based on the events whenever the bot is launched.			metadata a reg what configurations already exists in user's account and what is created by MontyCloud so while offboarding we won't change user's existing settings

<https://github.com/atlassian-internal/s...>Connect





This page has 1 GitHub link.

Connect to preview what's inside. Your teammates have connected this app.

Remediation	Parameters	Unsupp	BotMasterId	Owner	CreatedAt	UpdatedAt
-------------	------------	--------	-------------	-------	-----------	-----------

d			on	cript ion	on Type		orte d Regi ons			t	
	best practi ces for S3	Confi gRule Name	[task ref id]	confi grule desc	Day2 Task	1 { 2 "Trigg er type": Period ic, 3 "Input Param eters": {param Name: paramV alue}, 4 "Sourc e": {Sourc eIdent ifier} 5 "Scope ": {Compl ianceR esourc eTypes } 6 }	1 { 2 "R eg i o n s ": ["A f r i c a (C a p e T o w n) ", "E u r o p e (M i l a n) "] } 3 }		DAY2		
						1 { 2 "Ident ifier" : ACCESS _KEYS_ ROTATE D,					

 <https://github.com/atlassian-internal/s...> [Connect](#)



 This page has 1 GitHub link.

Connect to preview what's inside. Your teammates have connected this app.


```
21 ]
22 }
```

/bots/{botname}

Get

Details of the bot (Available in the registries)

```
1 {
2   "BotId": "BotId",
3   "Name": "<BotName>",
4   "Description": "
5   <Description>",
6   "Metadata": {
7     "Name": "
8     <BotName>",
9     "Description": "
10    <Description>",
11    "Avatar": "
12    <Avatar>"
13  },
14  "BotParameters": {
15    "InputParameters": [
16      {
17        "Name": "
18        Account"
19      },
20      {
21        "Name": "
22        Region"
23      },
24      {
25        "Name": "Run
26        Policy Interval",
27        "Values": [
28          "12hours"
29        ]
30      }
31    ]
32  },
33  "BotPolicies": [
34    {
35      "PolicyGroupName": "
36      <PolicyGroupName>",
37      "PolicyGroupDescription": "
38      <PolicyGroupDescription>",
39      "PolicyGroupId": "
40      <uuid(PolicyGroupName)>",
41      "Policies": [
42        {
43          "PolicyName": "
44          <PolicyName>",
45          "PolicyDescription": "
46          <PolicyDescription>",
47          "PolicyId": "
48          <uuid(PolicyName)>",
49          "InputType": "
50          <Text| TextArea|
51          List| CheckBox>"
52        }
53      ]
54    }
55  ],
56  "BotPermissions": {
```

 <https://github.com/atlassian-internal/s...> [Connect](#)

 This page has 1 GitHub link.

Connect to preview what's inside. Your teammates have connected this app.

				<pre> 42 "ResourceName": " 43 <ResourceName>", 44 "ResourceType": " 45 <ResourceType>" 46 }, 47 "BotNotifications": 48 [49 { 50 "NotificationType": " 51 <Slack Teams 52 PagerDuty>", 53 "WebHookURL": " 54 <WebHookURL>", 55 "Severity": " 56 <CRITICAL HIGH 57 MEDIUM>" 58 } 59] 60 }</pre>
/bots/:id/he alth	Get			
/bots/:id/co nfiguration s	Get	Available config of the selected bot		<pre> 1 { 2 "BotId": "BotId", 3 "RunPolicyInterval": 4 "<RunInterval>", 5 "ActionsNotifications": 6 [7 { 8 "NotificationType": " 9 <NotificationType>" 10 } 11] 12 }</pre>
/bots/:id/pr erequisites	Get			<pre> 1 { 2 "BotId": "BotId", 3 "Prerequisites": [4 "Check if AWS 5 Config is configured", 6 "Check if 7 delivery channel 8 exists", 9 "Check if 10 config recorder is 11 turned on" 12] 13 }</pre>
			<pre> 1 [2 { 3 "Region": " 4 <region>", 5 "AccountNumber": " 6 <AccountNumber>", 7 "DepartmentId": " 8 <DepartmentId>" 9 } 10]</pre>	<pre> 1 { 2 "BotId": "BotId", 3 "Target": { 4 "Account": " 5 <Account>", 6 "Region": " 7 <Region>" 8 }, 9 "PrerequisiteStatus": 10 "<Boolean>", 11 "Reasons": [</pre>


<https://github.com/atlassian-internal/s...>
Connect


 This page has 1 GitHub link.

Connect to preview what's inside. Your teammates have connected this app.

			<pre> 7] </pre>	<pre> 9 "Config rule limit has reached the limit" 10] 11 } </pre>
/bots/:id/configurations	Put	Update bot configurations	<pre> 1 { 2 "Interval": " <Interval>", 3 "ConfigRule": [], 4 5 "NotificationDetails": 6 [7 { 8 "NotificationType": " <NotificationType>", 9 "ChannelName": " <ChannelName>", 10 "Severity": [11 "<Severity>" 12], 13 "Metadata": {} 14 } 15] 16 } </pre>	
/bots/:id/insights/	Get	List insights		<pre> 1 { 2 "BotId": "BotId", 3 "Insights": [4 "Findings" 5] 6 } </pre>
/bots/:id/instances	Get	List instances of a selected bot		<pre> 1 { 2 "BotId": "<BotId>", 3 "Instances": [4 { 5 "BotInstanceId": "<BotInstanceId>", 6 "BotTarget": { 7 "Account": " <Account>", 8 "Region": " <Region>" 9 } 10 }, 11], 12 "CurrentState": " <CurrentState>" 13 } </pre>
 https://github.com/atlassian-internal/s... Connect  This page has 1 GitHub link. Connect to preview what's inside. Your teammates have connected this app.			<pre> 1 { 2 "Targets": [3 { 4 "Region": " <region>", 5 "AccountNumber": "<AccountNumber>", 6 "DepartmentId": " <DepartmentId>" 7 } 8] 9 } </pre>	<pre> 1 { 2 "Status": "<Status>", 3 "Message": " <Message>" 4 } </pre>

			<pre>8] 9 }</pre>	
/bots/:id/in stances	Delete	Offboard bot from selected targets	<pre>1 { 2 "Targets": [3 { 4 "Region": " <region>", 5 "AccountNumber": "<AccountNumber>", 6 "DepartmentId": " <DepartmentId>" 7 } 8] 9 }</pre>	<pre>1 { 2 "Status": "<Status>", 3 "Message": " <Message>" 4 }</pre>
/bots/:id/in stances/:id	Get	Details of the selected bot instance		<pre>1 { 2 "BotId": "BotId", 3 "<BotInstanceId>": { 4 "Account": " <Account>", 5 "Region": " <Region>" 6 } 7 }</pre>
/bots/:id/re mediate	POST	Trigger remediate	policyid, ruleid and remediation id	
/bots/:id/ap provals	Get			
/bots/appr ovals/:appr ovalId/appr ove? mc_token	Get	TBD		

Events:

Event	Data	Context	New Event	Description
	BotId		yes	When the Bot is getting deployed on the Day2 platform
	Bot Description			
	BotId		yes	When Bot's parameters are gets
	Bot's parameter list			

 <https://github.com/atlassian-internal/s...> [Connect](#)

 This page has 1 GitHub link.
Connect to preview what's inside. Your teammates have connected this app.

				registered on to the platform
bot.PrerequisitesRegistered	1. BotId 2. Bot's Prerequisite list		yes	When Bot's Prerequisite check list gets registered on to the platform
bot.InsightsDetailsRegistered	1. BotId 2. Bot's Insights details		yes	When Bot's Insights details gets registered to the platform
bot.InsightstypeAndActionMappingRegistered	1. BotId 2. Bot's Action mapping details		yes	When Bot's Action mapping gets registered to the platform (e.g. suppress finding)
bot.PolicyRuleRemediationMappingRegistered	1. BotId 2. Bot's Remediation Policy, Rule, Remediation and Remediation Type mapping		yes	When Bot's Remediation mapping gets registered to the platform
bot.RegisterCompleted	1. BotId		yes	When a Bot's registration gets completed
bot.DeploymentSucceeded	1. BotId		yes	When Bot gets successfully deployed
	BotId		yes	When Bot fails while deploying
	BotId departmentId accountNumber	1. CustomerId	yes	When prerequisite check completes for a target before launching a bot .


<https://github.com/atlassian-internal/s...>
[Connect](#)



This page has 1 GitHub link.

Connect to preview what's inside. Your teammates have connected this app.

	4. region 5. Prerequisite details			
bot.PrerequisiteFailed	1. BotId 2. DepartmentId 3. AccountNumber 4. region 5. Prerequisite details	1. CustomerId	yes	When prerequisite check fails for a target
bot.LaunchStarted	1. BotId 2. DepartmentId 3. AccountNumber 4. region	1. CustomerId	yes	When launching of a Bot starts within a Target
bot.InstanceDeployed	1. BotId 2. DepartmentId 3. AccountNumber 4. region	1. CustomerId	yes	When launching of a Bot gets completed within a Target
bot.DisablingStarted	1. BotId 2. DepartmentId 3. AccountNumber 4. region	1. CustomerId	yes	When off-boarding of a Bot gets started within a Target
bot.InstanceRemoved	1. BotId 2. DepartmentId 3. AccountNumber 4. region	1. CustomerId	yes	When off-boarding of a Bot gets completed within a Target


<https://github.com/atlassian-internal/s...>
Connect



This page has 1 GitHub link.

Connect to preview what's inside. Your teammates have connected this app.

bot.RemediationTriggered	1. BotId 2. DepartmentId 3. AccountNumber 4. region 5. Finding and Remediation details (Policy, Rule, Remediation Type)	1. CustomerId	yes	When Remediation gets triggered against findings.
bot.RemediationCompleted	1. BotId 2. DepartmentId 3. AccountNumber 4. region 5. Finding and Remediation details (Policy, Rule, Remediation Type)	1. CustomerId	yes	When Remediation gets completed.
bot.DataActionTriggered	1. BotId 2. DepartmentId 3. AccountNumber 4. region 5. Action Details	1. CustomerId	yes	When an Action is triggered for selected findings
	1. BotId 2. DepartmentId 3. AccountNumber 4. region 5. Action Details	1. CustomerId	yes	When an triggered Action gets completed for selected findings

×
<https://github.com/atlassian-internal/s...>
Connect

This page has 1 GitHub link.

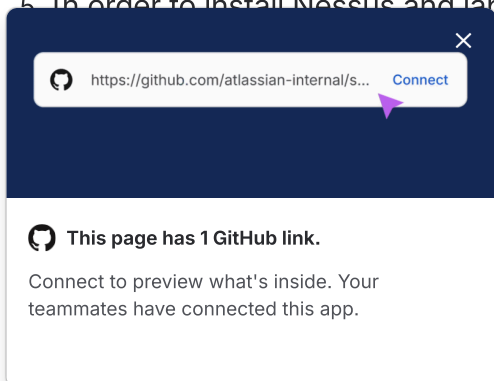
Connect to preview what's inside. Your teammates have connected this app.

bot.InsightsActionFailed	1. BotId 2. DepartmentId 3. AccountNumber 4. region 5. Action Details	1. CustomerId	yes	When an triggered Action fails for selected findings
--------------------------	---	---------------	-----	--

Open questions:

1. how to handle config rule limit exceeding gracefully?
 - a. we calculate config rules deployed in account/region and based on user's config rule selection notify user about exceeding limit.
 - b. we let user deploy config rules till exception occurs for limit exceed.
2. how to handle region specific config rule deployment?
 - a. we need to store metadata about each config rule with list of non-supported regions.
downside: if in future AWS provides support to these regions then we would need to update our registry too.
3. While deploying “ec2-managedinstance-applications-required” config rule we will have to enter the application names with versions as parameters for the rule, but each customer might have different application versions how do we handle this..?
suggestion: we can ignore versions for MVP. Just check for application names.
4. For remediation actions in DAY1 solution the automation documents have AWS approve as a step which expects us to pass an IAM user as an input parameter, we do not discover IAM users as of now so how will we be handling this.

5. In order to install Nessus and Iaps agents on servers, we will need installation keys and both even check if these applications exist as they are very specific the day1 team has built..?
for SSM agent and Cloudwatch agent for MVP..?
approval flow for remediation..? The automation approval flow
input parameters for the task and as of now we don't discover



7. Should we send compliance notifications for all the config rules to slack..?
(NOTE: it could be noisy to user)
8. what notifications should we publish non-compliant, compliant or both..?
9. what will happen to bots when customer account is offboarded from platform?

